

architecture is that it offers device options that can compete with 8/16-bit processors and high-end 32- and 64-bit variations with single and multiple cores. This offers multiple options with a single open architecture. One question as laymen we get is, why a processor has become the critical issue now?

Three critical aspects are important for any country to maintain independence and progress on the technology front. These are:

1. Independence to decide and develop technologies that are a must for the country's progress.

2. Availability of parts (no supply chain bottlenecks) freely. We should avoid the current situation of companies waiting for ICs and other parts.

3. With the world becoming connected and proliferation of connected devices, one of the worrisome aspects is the security. We are not talking about the phishing and hacking but the connected devices being hijacked and spoofed to impact the day-to-day life of the citizens. One example would be, as part of smart city project there will be sensors which monitor the environment—especially pollution. Imagine if a hacker replaces the installed (good/genuine) devices with faulty devices so that even when the environment is polluted the devices report everything is fine. You can understand the impact on the society as a result.

This essentially means that the communication must be encrypted and, above all, the devices have to be validated and approved by the installer. While this looks easy, implementation is a challenge which needs a solution that mandates the devices to be uniquely identified (yes, each IC should have a unique certified ID!). In addition, these devices should be validated before they are allowed into service, and need to be periodically checked for presence of a valid device.

IIT Madras RISC-V projects

SHAKTI is an open source initiative by the Reconfigurable Intelligent Systems Engineering (RISE) group at Indian Institute of Technology, Madras to develop the first indigenous Indian industrial-grade processor. The aim of SHAKTI initiative includes building an open source production-grade processor, complete systems on chips (SoCs), development boards, and SHAKTI based software platform.

The primary focus of the team is architecture research to develop SoCs, which are competitive with commercial offerings in the market concerning area, power, and performance. All the source codes for SHAKTI are open-sourced under the Modified BSD License. The project was funded by Ministry of Electronics and Information Technology (MeITY), government of India.

Two tapeouts (ICs fabricated and tested) of the C-class processors have been performed. They have been codenamed as RIMO and RISECREEK (test chip).

RIMO is the code name of the SHAKTI C-class based SoC that has been taped-out at Semi-Conductor Laboratory (SCL) at Chandigarh using 180nm process technology. The 144sq mm chip has been tested to operate at a frequency of up to 70MHz. The chip has been packaged on a 208-pin ceramic quad flat pack (CQFP).

RISECREEK is the code name of the SHAKTI C-class based SoC that has been taped-out at Intel's Oregon fab using a 22nm FinFET process. The 16mm² chip has been tested to operate at a frequency of up to 350MHz. The chip has been packaged on a 208-pin ball grid array (BGA).

MOUSHIK is the code name of the SHAKTI E-class based SoC that has been taped-out at Semi-Conductor Laboratory (SCL) at Chandigarh using 180nm process technology. It operates at 100MHz and was developed along with a motherboard called Ardonyx 1.0.

We can see the evolving situation demands that the processors cannot be a plain vanilla type but a device whose history can be traced and identified. This requirement is going to be implemented as we go forward. USA NIST has released NIST 8228 (Considerations for Managing Internet of Things (IoT) Cybersecurity and Privacy Risks) standard for this purpose.

Incidentally, in the Indian context, UIDAI has already updated its next-generation fingerprint sensor's compliance to secure processors and to be independently verified and validated by STQC before the sensors are put in the field. This ensures absolute safety of the customer's data. The same is true for military equipment too.

The situation in India

Real purpose of this article is to educate readers on what it takes to design a processor in India and persuade companies and other organisations to use it. Let us first see the lifecycle of a semiconductor design, which would help us under-

stand where we stand today.

Fig. 1 shows very simple flow chart of an IC design for easy understanding. The lifecycle consists of three major blocks:

1. Design phase, which has two steps before the design is ready for the IC's fabrication. This is purely an engineering effort that requires experienced designers who can develop an IC using languages like Verilog or VHDL. The process of coding an IC using one of these languages is typically called front-end design.

Typically, a processor design involves the coding of the main core and integration of libraries for different peripherals needed in the processor/controller. In the next step, the coded IC has to be tested completely using test benches. These are Verilog/VHDL codes written exclusively for testing (something very close to simulation of the boards for their working before committing to the PCB).

2. The tested code is converted into the GDSII format, which is used by the IC fabricators—also